Math 299 Exercises

Lance Remigio

October 9, 2025

Contents

1 Week 1

1.1 Zorn's Lemma

Problem 1 (4.1.2)**.** Let X be the set of all real-valued functions on X on the interval $[0, 1]$, and let $x \leqq y$ to mean that $x(t) \leq y(t)$ for all $t \in [0, 1]$. Show that this defines a partial ordering. Is it total ordering? Does X have a maximal elements?

Proof. Define $F = \{\text{all real-valued functions on the interval } [0, 1]\}$. First, we show reflexivity. Let $x \in F$. Then for all $t \in [0, 1]$, $x(t) \leq x(t)$. Hence, $x \leqq x$ and so F contains the reflexive property. For antisymmetry, suppose $x \leqq y$ and $y \leqq x$. Then for all $t \in [0, 1]$, we have $x(t) \leq y(t)$ and $y(t) \leq x(t)$. Then $x(t) = y(t)$ (using the partial ordering of the real numbers) for all $t \in [0, 1]$. Then $x = y$ and so antisymmetry is satisfied. For transitivity, suppose $x \leqq y$ and $y \leqq z$. Then $x(t) \leq y(t)$ and $y(t) \leq z(t)$ for all $t \in [0, 1]$. Then $x(t) \leq z(t)$ for all $t \in [0, 1]$ using the transitivity of the real numbers. Hence, $x \leqq z$ and so transitivity is satisfied.

F is totally ordered, but has no maximal elements. ■

Problem 2 (4.1.5)**.** Prove that a finite partially ordered set A has at least one maximal element.

Proof. ■

1.2 Hahn-Banach Theorem

Problem 3 (Existence of a Sublinear Functional)**.** Show that a sublinear functionnal p satisfies $p(0) = 0$ and $p(-x) \geq -p(x)$.

Proof. Define $P : X \to \mathbb{R}$ by

$$p(x) = \|x\|$$

where X is some vector space. Then for $x = 0$, we have

$$p(0) = \|0\| = 0.$$

Note that for all $x \in X$, we have

$$p(-x) = \| - x\| = | - 1|\|x\| = 1 \cdot \|x\| = \|x\| > 0$$

and $p(-x) = p(x)$. Now, note that for all $x \in X$

$$-\|x\| \leq \|x\| \iff -p(x) \leq p(x) = p(-x).$$

Thus, for all $x \in X$, $p(-x) \geq -p(x)$. ■

Problem 4 (Convex Set). If p is a sublinear functional on a vector space X, show that $M = \{x : p(x) \leq \gamma, \gamma > 0 \text{ fixed}\}$, is a convex set.

Proof. Assume that p is a sublinear functional on a vector space X. Let $W = \{v = \alpha y + (1-\alpha)z : 0 \leq \alpha \leq 1\}$. Our goal is to show that $W \subseteq M$. To this end, let $v \in W$ be arbitrary. Then for some $y, z \in M$, we have $v = \alpha y + (1-\alpha)z$ where $\alpha \in \mathbb{R}$. Since $y, z \in M$, we have $p(y) \leq \gamma$ and $p(z) \leq \gamma$ where $\gamma > 0$ is fixed. Our goal is to show that $p(v) \leq \gamma$ for fixed γ. Using the sublinearity of p, we get that

$$\begin{aligned} p(v) &= p(\alpha y + (1-\alpha)z) \\ &\leq p(\alpha y) + p((1-\alpha)z) \\ &= |\alpha| p(y) + |(1-\alpha)| p(z) \\ &= \alpha p(y) + (1-\alpha)p(z) && (0 \leq \alpha \leq 1) \\ &\leq \alpha\gamma + (1-\alpha)\gamma \\ &= \gamma. \end{aligned}$$

Hence, we conclude that $W \subseteq M$ since v was an arbitrary element and so M is a convex set. ■

Problem 5. Let p be a sublinear functinoal on a real vector space X. Let f be defined on $Z = \{x \in X : x = \alpha x_0,\ \alpha \in \mathbb{R}\}$ by $f(x) = \alpha p(x_0)$ with fixed $x_0 \in X$. Show that f is a linear functional on Z satisfying $f(x) \leq p(x)$.

Proof. First, we will show that f is a linear functional on Z. Let $u, v \in Z$. Then $u = \alpha_1 x_0$ and $v = \alpha_2 x_0$ for $\alpha_1, \alpha_2 \in \mathbb{R}$. Let $\delta \in \mathbb{R}$. Observe that

$$f(x) = \alpha p(x_0) = p(\alpha x_0).$$

Using this observation, we have

$$\begin{aligned} \delta f(u) + f(v) &= \delta\alpha_1 p(x_0) + \alpha_2 p(x_0) \\ &= (\delta\alpha_1 + \alpha_2)p(x_0) \\ &= p(\delta\alpha_1 x_0 + \alpha_2 x_0) \\ &= p(\delta u + v) \\ &= f(\delta u + v). \end{aligned}$$

Hence, f is a linear functional. Using our observation again, we can also see that for $\alpha > 0$, $f(x) = p(x)$ and so $f(x) \leq p(x)$ for all $x \in Z$. Clearly, the inequality holds if $\alpha = 0$. Now, suppose $\alpha < 0$. Then

$$f(x) = \alpha p(x_0) = |\alpha| p(x_0) = p(|\alpha| x_0) = p(x)$$

and so $f(x) \leq p(x)$ for all $x \in Z$. ■

Problem 6. If p is a sublinear on a real vector space X, show that there exists a linear functional $\tilde{f}$ on X such that

$$-p(-x) \leq \tilde{f}(x) \leq p(x).$$

Proof. Define Z as in the set in the previous problem and define $f(x) = \alpha p(x_0)$ with fixed $x_0 \in X$. Using the same problem, we proved that f defines linear functional such that $f(x) \leq p(x)$ for all $x \in Z$. Since $Z \subseteq X$, we can find an extension (via the Hahn-Banach Theorem) $\tilde{f}$ that is also a linear functional from

Z to X satisfying $\tilde{f}(x) \leq p(x)$ for all $x \in X$. All we need to show now is $-p(-x) \leq \tilde{f}(x)$. Since the bound in the previous statement holds for all $x \in X$, we have

$$-\tilde{f}(x) = \tilde{f}(-x) \leq p(-x).$$

Multiplying through by a negative, we now have

$$\tilde{f}(x) \geq -p(-x)$$

which completes our proof. ■

2 Week 7: Reflexive Spaces

Theorem (Dual Space (2.10-4)). The dual space X' of a normed space X is a Banach Space (whether or not X is).

Proposition. The dual space of $\mathbb{R}^n$ is $\mathbb{R}^n$.

Theorem (3.8-1). Every bounded linear functional f on a Hilbert space H can be represented in terms of the inner product , namely,

$$f(x) = \langle x, z \rangle$$

where z depends on f, is uniquely determined by f and has norm

$$\|z\| = \|f\|.$$

Recall the definition of Algebraic Reflexivity of a vector space X that was introduced in section 2.8.

Definition (Algebraic Reflexivity). A vector space X is said to be **algebraic reflexive** if the canonical mapping $C : X \to X^{**}$ is surjective.

- Here we see that X^{**} is the second algebraic dual space of X and the mapping C is defined by $x \mapsto g_x$ where

  $$g_x(f) = f(x). \quad ; \qquad ((1) \quad (f \in X^*))$$

  that is, for any $x \in X$ the image is the linear functional g_x defined by the above.
- If X is finite-dimensional, then X is algebraic reflexive.

Consider a normed space X and its dual space X' (the space of all bounded linear functionals). Observe that $(X')'$ is the **second dual space** of X (or **bidual space** of X). We define a similar mapping to the above statement, by setting

$$g_x(f) = f(x) \quad . \qquad ((2), \quad (f \in X'))$$

The only difference here is that f is now bounded and so g_x is bounded.

Lemma (Norm of g_x). For every fixed x in a normed space X, the functional g_x defined above is a bounded linear functional on X', so that $g_x \in X''$, and has the norm

$$\|g_x\| = \|x\|. \qquad (3)$$

Proof. It is easy to see that g_x is linear, and (3) follows from (2) and corollary 4.3-4; that is,

$$\|g_x\| = \sup_{\substack{f \in X' \\ f \neq 0}} \frac{|g_x(f)|}{\|f\|} = \sup_{\substack{f \in X' \\ f \neq 0}} \frac{|f(x)|}{\|f\|} = \|x\|. \qquad (4)$$

■

- This lemma tells us that for every $x \in X$, there corresponds a unique bounded linear functional $g_x \in X''$ given by (2) which defines the **canonical mapping** of X into X'':

$$\begin{aligned} C : X &\longrightarrow X'' \\ x &\longmapsto g_x \end{aligned} \tag{5}$$

- This map is linear and injective and preserves the norm; that is, in terms of mapping the domain to its range, it will be an isomorphism of normed spaces.
- Roughly speaking, this allows us to embed any normed space X into the bidual X'' "faithfully", preserving the geometry of the space. That is, every normed space X is a subset of X".
- This lemma gives us one more step towards our notion of reflexivity. All we need is for our canonical map to be surjective which will be outlined below.
- Conceptually, we can interpret this result as connecting local evaluations $f(x)$ to the "global structure" of X".

> **Lemma** (Canonical Mapping). The canonical mapping C given by (5) is an isomorphism of the normed space X onto the normed space $\mathcal{R}(C)$, the range of C.

Proof. We can see easily see the linearity of C. Indeed, we have

$$g_{\alpha x+\beta y}(f) = f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) = \alpha g_x(f) + \beta g_y(f).$$

In particular, we have $g_x - g_y = g_{x-y}$. Thus, we obtain

$$\|g_x - g_y\| = \|g_{x-y}\| = \|x - y\|.$$

This tells us that C is an isometry and so it must preserve the norm. Because isometries are injective, C is also injective. This can be directly seen from the above formula. Indeed, for any $x \neq y$, we have $g_x \neq g_y$ by axiom (N2) in section 2.2. Since we are considering maps from X to the range of C, it follows that C is also surjective. Thus, C must be bijective. ■

> **Definition** (Embeddable). We say that a set X is **embeddable** in a normed space Z if X is isomorphic with a subspace of Z.

- From our Canonical Mapping Lemma, we can see that X is embeddable in X'' since the range of C is a subspace of X'' and it is isomorphic to X.
- The mapping C is called the **canonical embedding** in this case.
- In general, the range of C will be a proper subspace of X''.
- If the range of C happens to be a surjection, then we will call it a different name which will be defined below.

> **Definition** (Reflexivity). A normed space X is said to be **reflexive** if
>
> $$\mathcal{R}(C) = X''$$
>
> where $C : X \to X''$ is the canonical mapping given by (5) and (2).

- If X is reflexive, it is isomorphic (and hence isometric) with X'' by Lemma 4.6-2.
- However, the converse does not hold.
- Completeness does not necessarily imply reflexivity.
- However, the converse does hold which will be outlined below.
- Conceptually, this lemma allows us to say that X and X" are the same as Banach spaces.

Theorem (Completeness)**.** If a normed space X is reflexive, it is complete (hence a Banach space).

Proof. Since X'' is the dual space of X', it is complete by Theorem 2.10-4. Reflexivity of X means that $\mathcal{R}(C) = X''$. Completeness of X now follows from that of X'' by Lemma 4.6-2. ■

- An example of a reflexive space is $\mathbb{R}^n$ which follows from example 2.10-5.
- For finite dimensions, every linear functional on a general normed space X is bounded and so $X' = X^*$ which implies further that
- Reflexivity allows for infinite dimensional analysis manageable; that is, it restores many of the compactness and duality properties that finite-dimensional analysis so powerful.

Theorem (Finite Dimension)**.** Every finite-dimensional normed space is reflexive.

- l^p with $1 < p < +\infty$ is reflexive.
- $L^p[a, b]$ with $1 < p < +\infty$ is reflexive.
- Non-examples are $C[a, b]$, l^1, $L^1[a, b]$, l^∞ and the subspaces c and c_0 of l^∞ where c is the space of all convergent sequence of scalars and c_0 is the space of all sequences of scalars converging to zero.

Theorem (Hilbert Space)**.** Every Hilbert Space H is reflexive.

Proof. Our goal is to show that the canonical mapping $C : H \to H''$ is surjective by showing that for every $g \in H''$, there exists an $x \in H$ such that $g = Cx$. Define $A : H' \to H$ by $Af = z$, where z is given the Riesz Representation $f(x) = \langle x, z\rangle$. From this, we see that A is bijective and isometric. Note that A is conjugate linear, as we see from (16) from section 4.5. Now, H' is complete by 2.10-4 and a Hilbert space with inner product defined by

$$\langle f_1, f_2\rangle_1 = \langle Af_2, Af_1\rangle.$$

It can be easily shown that $\langle , \rangle_1$ is an inner product on H.

Let $g \in H''$ be arbitrary. Let its Riesz Representation be

$$g(f) = \langle f, f_0\rangle_1 = \langle Af_0, Af\rangle.$$

We now remember that $f(x) = \langle x, z\rangle$ where $z = Af$. Writing $Af_0 = x$, we have

$$\langle Af_0, Af\rangle = \langle x, z\rangle = f(x).$$

Hence, we have $g(f) = f(x)$ and so $g = Cx$ by definition of C. Hence, C is surjective and so H must be reflexive. ■

- Separability of the dual of a normed space implies separability of X (note the converse is not generally true).
- If a normed space X is reflexive, then X'' is isomorphic with X by 4.6-2, and so X implies separability of X''. Thus, the dual space X' must also be separable.
- *A separable normed space X with a nonseparable dual space X' cannot be reflexive.*

Example (l^1 is not reflexive)**.** Note that l^1 is separable by 1.3-10, but the dual of l^1 which is l^∞ is not via 2.10-6 and 1.3-9.

Lemma (Existence of a Functional). Let Y be a proper closed subspace of a normed space X. Let $x_0 \in X \setminus Y$ be arbitrary and

$$\delta = \inf_{\tilde{y} \in Y} \|\tilde{y} - x_0\| \tag{6}$$

the distance from x_0 to Y. Then there exists an $\tilde{f} \in X'$ such that

$$\|\tilde{f}\| = 1 \qquad \tilde{f}(y) = 0 \text{ for all } y \in Y, \qquad \tilde{f}(x_0) = \delta. \tag{7}$$

Proof. Consider the subspace $Z \subset X$ spanned by Y and x_0 and define a bounded linear function f on Z by

$$f(z) = f(y + \alpha x_0) = \alpha\delta. \qquad (y \in Y)$$

where $z \in Z = \operatorname{span}(Y \cup \{x_0\})$ has a unique representation

$$z = y + \alpha x_0 \qquad y \in Y. \tag{8}$$

It is relatively easy to see that f is linear. Furthermore, because Y is a closed set and $\delta > 0$, we have $f \neq 0$. We may consider a few cases for α. If $\alpha = 0$, then

$$f(y) = f(y + 0x_0) = 0 \cdot \delta = 0.$$

If $\alpha = 1$ and $y = 0$, then we have $f(x_0) = \delta$. Now we show that f is bounded. If $\alpha = 0$, then $f(z) = 0$. If $\alpha \neq 0$, then since Y is a subspace of X, we know that $-(1/\alpha)y \in Y$. Using (6), we obtain the following inequality

$$\begin{aligned} |f(z)| = |\alpha|\delta = |\alpha| \inf_{\tilde{y} \in Y} \|\tilde{y} - x_0\| \\ &\leq |\alpha| \| -\frac{1}{\alpha} y - x_0\| \\ &= \|y + \alpha x_0\|, \end{aligned}$$

that is, $|f(z)| \leq \|z\|$. Thus, we see that f is bounded and so

$$\frac{|f(z)|}{\|z\|} \leq 1 \Longrightarrow \|f\| \leq 1.$$

Next we show that $\|f\| \geq 1$. Using the definition of the infimum, Y contains a sequence (y_n) such that $\|y_n - x_0\| \to \delta$. Let $z_n = y_n - x_0$. Then we have $f(z_n) = -\delta$ by (8) with $\alpha = -1$. Also

$$\|f\| = \sup_{\substack{z \in Z \\ z \neq 0}} \frac{|f(z)|}{\|z\|} \geq \frac{|f(z_n)|}{\|z_n\|} = \frac{\delta}{\|z_n\|} \underset{n \to \infty}{\longrightarrow} \frac{\delta}{\delta} = 1.$$

Hence, $\|f\| \geq 1$ and so we conclude that $\|f\| = 1$. Using the Hahn-Banach theorem for normed spaces we can extend f to X while preserving its norm. ■

Theorem (Separability). If the dual space X' of a normed space X is separable, then X itself is separable.

Proof. We assume that X' is separable. Then the unit sphere

$$U' = \{f : \|f\| = 1\} \subseteq X'$$

also contains a countable dense subset, say, (f_n). Since $f_n \in U'$, we have

$$\|f_n\| = \sup_{\|x\|=1} \|f_n(x)\| = 1.$$

By the definition of the supremum, we can find points $x_n \in X$ of norm 1 such that

$$\|f_n(x_n)\| \geq \frac{1}{2}.$$

Let Y be the closure of $\operatorname{span}(\{x_n\})$. Then Y is separable because Y has a countable dense subset, namely, the set of all linear combinations of x_n's with coefficients whose real and imaginary parts are rational.

Our goal is to show that $Y = X$. Suppose $Y \neq X$. Then, since Y is closed, by lemma 4.6-7 there exists an $\tilde{f} \in X'$ with $\|\tilde{f}\| = 1$ and $\tilde{f}(y) = 0$ for all $y \in Y$. Since $x_n \in Y$, we have $\tilde{f}(x_n) = 0$ and for all n,

$$\begin{aligned} \frac{1}{2} \leq |f_n(x_n)| &= |f_n(x_n) - \tilde{f}(x_n)| \\ &= |(f_n - \tilde{f})(x_n)| \\ &\leq \|f_n - f\| \|x_n\| \end{aligned}$$

where $\|x_n\| = 1$. Hence, $\|f_n - f\| \geq \frac{1}{2}$, but this contradicts the assumption that (f_n) is dense in U' because $\tilde{f}$ is itself in U' where $\|\tilde{f}\| = 1$. ■